

LAB 12 REPORT

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Lab Title: Sets & Dictionary

Theory: Sets are unordered collections of unique values in Python. They do not allow duplicate elements and are used for storing distinct data.

Dictionary stores data in the form of **key-value pairs**, where each key is used to access its related value.

TASKS

```
# TASK 1
s={"ce","cs","ibm","ce"}
for i in s:
    if i=="ce":
        print(i)
```

```
#TASK 2
l={1,2,3,4,5,6,7,8,9}
l.add("ae")
print(l)
```

```
#TASK 3
j={2,1,4,3,7,3}
j.discard(2)
print(j)
```

```
#TASK 4
j={2,1,4,3,7,3}
j.discard(9)
print(j)
```

```
#TASK 5
j={2,1,4,3,7,3}
j.remove(2)
print(j)
```

```
#TASK 6
j={2,1,4,3,7,3}
```

```
j.remove(7)
print(j)
```

Output:

```
ce
{1, 2, 3, 4, 5, 6, 7, 8, 9, 'ae'}
{1, 3, 4, 7}
{1, 2, 3, 4, 7}
{1, 3, 4, 7}
{1, 2, 3, 4}

Process finished with exit code 0
```

TASKS

```
#TASK 7
k={2,3,4,5,6,7,8,9}
k.pop()
print(k)
```

```
#TASK 8
i1={"muhammad sohaib"}
i2={"qaisrani"}
i3={"dgk"}
a=set.union(i1, i2,i3)
print(a)
```

```
#TASK 9
i1={"muhammad sohaib"}
i2={"qaisrani","dgk"}
i3={"dgk"}
a=set.union(i1, i2,i3)
print(a)
```

```
#TASK 10
s1={1,2,3,"sohaib"}
s2={4,5,6,"dgk"}
s1.update(s2)
print(s1)
print(s2)
```

Output:

```
{3, 4, 5, 6, 7, 8, 9}
{'dgk', 'qaisrani', 'muhammad sohaib'}
{'dgk', 'qaisrani', 'muhammad sohaib'}
{1, 2, 3, 4, 5, 6, 'dgk', 'sohaib'}
{'dgk', 4, 5, 6}

Process finished with exit code 0
```

DICTIONARY

TASK 1

```
student = {"name": "Ali", "marks": 85, "city": "Lahore"}
```

```
for key, value in student.items():
    print(key, value)
```

```
student["marks"] = 90
print(student)
```

Output:

```
name Ali
marks 85
city Lahore
{'name': 'Ali', 'marks': 90, 'city': 'Lahore'}

Process finished with exit code 0
```

TASK 2

```
products = {"Laptop": 80000, "Mobile": 50000, "Tablet": 30000}
products["Mouse"] = 5000
for p, price in products.items():
    if price > 40000:
        print(p, price)
```

Output:

```
Laptop 80000
```

```
Mobile 50000
```

```
Process finished with exit code 0
```